Graduation Guide

MSc Geomatics for the Built Environment

v2026–2027

Faculty of Architecture and the Built Environment
Delft University of Technology

Preface

The graduation phase is the last step in the MSc Geomatics for the Built Environment programme. It is completed in the form of a 45 ECTS individual research project (course GEO2022).

The topic of the scientific project can be related to any of the subjects offered in the MSc Geomatics for the Built Environment programme.

The project can be performed in cooperation with a company/institution, which provides a use case or a problem statement, but the work is always supervised by two scientific staff members of the university, one of whom is part of the MSc Geomatics programme (see Chapter 2).

More information, news, updates, downloadable templates, and a calendar can be found on the official website: https://geomatics.bk.tudelft.nl/geo2022/

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1. Overview of the graduation

1.1. Structure and duration

The *Geomatics Graduation Studio* (the graduation phase where the MSc thesis is written; GEO2022) is 45 ECTS and generally lasts three quarters.

It normally takes place during the last three quarters of the second Master's year (quarters Q6+Q7+Q8), but it can be started at the beginning of any quarter.

Note, however, that the thesis topic selection already takes place in Q5 (to ensure that the work starts on the first day of Q6), and the Finalisation presentation might take place in week 5.1, the first week of the summer period after Q8 (the week numbering follows the calendar in Chapter 1.3; see the standard planning in Appendix A).



1.2. Admission and milestones

To be admitted to GEO2022, students have to register for the Kick-off. To register for the Kick-off, the student must have obtained 45 ECTS from 9 out of the 10 mandatory courses, and have obtained 15 out of the 25 ECTS from the electives by the time of the Kick-off.

The graduation phase is divided into the following milestones:

	Milestone	Details
0	Preparation	Selection of the responsible supervisor and of a thesis topic, either by picking from a topic from the list or by agreeing on a custom topic.
1	Kick-off	'Go/no-go' assessment of the graduation plan; 15-min presentation, 15-min questions and discussion.
2	Midterm	Structured meeting with both supervisors to discuss progress and updated plans. The form is decided by the supervisors.
3	Green-light	Assessment of the complete draft MSc thesis; 20-min presentation, 20-min questions and discussion.
4	Finalisation	Assessment of the final MSc thesis; 20-min presentation, 25-min questions and discussion, diploma signing.

1.3. Calendar

All assessments (except Midterm) should be held in the weeks indicated below. Please be aware that there are deadlines for the registration for each assessment. For the academic year 2026–2027, this implies the following weeks and deadlines, as registered in the Geomatics graduation calendar:

Academic Calendar 2026/2027 — **Graduation**

AUTUMN SEMESTER

Calendar Week	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	1	2	3	4
Teaching week	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10	2.1	2.2	2.3	2.4	2.5	2.6	Christmas period		2.7	2.8	2.9	2.10
	Aug.	Sept.				Oct.				Nov.					Dec.				Jan.			
Monday	31	7	14	21	28	5	12	19	26	2	9	16	23	30	7	14	21	28	4	11	18	25
Tuesday	1	8	15	22	29	6	13	20	27	3	10	17	24	1	8	15	22	29	5	12	19	26
Wednesday	2	9	16	23	30	7	14	21	28	4	11	18	25	2	9	16	23	30	6	13	20	27
Thursday	3	10	17	24	1	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14	21	28
Friday	4	11	18	25	2	9	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22	29

SPRING SEMESTER

Calendar Week	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Teaching week	Spring break	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	3.10	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	4.10
	Feb.				Mar.					Apr.					May				June		
Monday	1	8	15	22	1	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14	21
Tuesday	2	9	16	23	2	9	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22
Wednesday	3	10	17	24	3	10	17	24	31	7	14	21	28	5	12	19	26	2	9	16	23
Thursday	4	11	18	25	4	11	18	25	1	8	15	22	29	6	13	20	27	3	10	17	24
Friday	5	12	19	26	5	12	19	26	2	9	16	23	30	7	14	21	28	4	11	18	25

SUMMER PERIOD

Calendar Week	26	27	28	29	30	31	32	33	34
Summer period	5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9
	June	July				Aug.			
Monday	28	5	12	19	26	2	9	16	23
Tuesday	29	6	13	20	27	3	10	17	24
Wednesday	30	7	14	21	28	4	11	18	25
Thursday	1	8	15	22	29	5	12	19	26
Friday	2	9	16	23	30	6	13	20	27

PUBLIC HOLIDAYS

- **Christmas period** 21 Dec – 1 Jan
- **Spring break** 1 Feb – 5 Feb
- **Good Friday** 26 March
- **Easter** 28 Mar – 29 Mar
- **King's Day** 27 April
- **Liberation Day** 5 May
- **Ascension Day** 6 May
- **Free day** 7 May
- **Whit Monday** 17 May

DAY COLOURS LEGEND

- Deadline to submit thesis topic
- Final registration date for Green-light and Finalisation assessments
- Final registration date for Kick-off assessments

WEEK COLOURS LEGEND

- Education
- No education
- Kick-off assessments
- Midterm assessments
- Green-light assessments
- Finalisation assessments

1.4. Retakes

All assessments are offered in every quarter.

At the Kick-off, the supervisors determine whether the student's plan and presentation meet the requirements, resulting in a 'go' to the Midterm milestone. If a 'no-go' is given, the Kick-off can be done again in the next quarter.

The same applies to the Green-light, where the supervisors give a 'go' to proceed to the public Finalisation at least 4 weeks later; if the requirements are not met, the consequent 'no-go' results in a retake of the Green-light in the next quarter.

1.5. Scheduling and registration

The Kick-off and Green-light assessments are scheduled by the responsible supervisor in the GEO2022 website. The Finalisation is scheduled by the responsible supervisor via the *SuperSaaS* tool; if not available, by email to graduation-bk@tudelft.nl. All assessments are scheduled in the standard time slots (see Appendix C), before the deadlines in the Geomatics graduation calendar. This makes it possible to reserve a room for the assessment, and gives the student, supervisors, co-reader and delegate time to record the appointment in their agendas. If the reserved time slot needs to be changed, this should be done by communicating with the coordinator of the Geomatics Graduation Studio (for the Kick-off and Green-light) and by email to graduation-bk@tudelft.nl (for the Finalisation).

For registering the graduation process, the faculty uses the *MyCase* registration tool (see Appendix A). It is the student's task to start their own 'graduation case'. MyCase is used for the registration of assessment dates, submission of the deliverables before an assessment, registration of the assessments, and feedback. Moreover, the names of the supervisors, co-reader and delegate, and the progress and assessment results are registered in this application. After each assessment, the responsible supervisor completes the assessment (see Chapter 4 and Appendix E).

1.6. Final grades and cum laude

The grades for the graduation project (GEO2022) are decided at the Finalisation milestone by the two supervisors and the co-reader: the supervisors assess the research, process, and communication criteria of the rubric, while the co-reader assesses the research and communication criteria only (see Appendix E and Chapter 2).

Based on the results gained for the courses in the Master's Programme and the final thesis, the student can be awarded a cum laude distinction. A student will receive this if all of the following requirements are met:

1. The weighted average of the grades of the courses, not including the graduation project, must be at least 8.0; only the 120 ECTS of the programme are used in the calculation, and passes ("V" in OSIRIS) and exemptions ("VR" in OSIRIS) are not taken into consideration;
2. No more than 20 ECTS in total may be completed with a pass ("V" in OSIRIS) or an exemption ("VR" in OSIRIS); courses taken during an exchange semester and assessed as pass/fail are not counted towards this maximum;
3. No courses may be completed after a repair resulting in a 'pass';
4. The final grade of the graduation project must be at least 8.5;

5. The study duration cannot exceed the nominal study duration plus one semester, taking into account study delays covered by the TU Delft Student Support Fund Scheme.

For students who do not meet all criteria above, especially the fifth, for instance because they were enrolled as a student in a specific semester but did not participate in any course or examination in that semester, an exemption may apply. This should be arranged before the Green-light application date (by sending a request by email to the Board of Examiners: BoardofExaminers-BK@tudelft.nl).

1.7. Ethics

In projects involving humans, certain types of data processing increase the risks to the human participants. Planning such projects requires additional evaluations and advice from university staff before ethical approval can be received and the project can begin. In the case of a graduation project, obtaining additional advice or permits may delay the project with an extra semester. Therefore, all students have to check their risk, by completing the Data management checklist (Appendix G) before the Kick-off. Only if the graduation project involves working with data from human participants, should the rest of the checklist be completed and contact be made with the Human Research Ethics Committee (HREC).

Completing the checklist is recorded in MyCase as part of the Kick-off (see Appendix A). A ‘no’ answer to the human-participation question is accepted at the Kick-off; the answers can still be changed and the HREC approval letter uploaded up to and including the Green-light. Note that the research itself can only begin once the HREC approval has been received.

2. Supervisory team

The student's supervisory team is composed of:

1. one responsible supervisor;
2. one second supervisor;
3. one co-reader.

2.1. Responsible supervisor

The responsible supervisor (also called first or main supervisor) is an expert in the graduation project field and must be a staff member of the MSc Geomatics for the Built Environment and must have the UTQ diploma (University Teaching Qualification).

Students choose this supervisor during the Preparation milestone, either by directly contacting them to agree on a custom topic or by selecting a predefined topic linked to the supervisor. The responsible supervisor is also responsible for the MyCase administration and for inviting the student, the second supervisor, and the co-reader to the assessment sessions (see Appendix B).

2.2. Second supervisor

The second supervisor is a scientific staff member from TU Delft whose expertise complements that of the responsible supervisor. They may be predefined based on the selected topic or chosen by the responsible supervisor with input from the student. Ideally, the second supervisor is determined at the Preparation phase and must be confirmed before the Kick-off.

2.3. Co-reader

The co-reader is a scientific staff member of TU Delft or another university who is an expert in the field of the graduation project. Their role is to independently assess the quality of the student's work in an unbiased way and to help with marking at the Finalisation. Because they are only involved at the end of the graduation, the co-reader does not assess the process: they contribute to assessing the research and communication criteria of the rubric (see Appendix E), while the process criterion is only assessed by the supervisors. If both supervisors are from the same department, the co-reader should preferably not be part of the same department. The co-reader is chosen by the supervisors in collaboration with the student. The co-reader is preferably defined at the Preparation and must be defined before the Finalisation.

2.4. Delegate of the Board of Examiners

Since the Board of Examiners is responsible for ensuring the quality of all examinations, a delegate is appointed by the Board of Examiners of the Faculty before the Kick-off: a staff member of the Faculty of Architecture and Built Environment who is familiar with the graduation process. If both supervisors are from the same department, the delegate must be from a different one. The delegate functions independently from the supervisors and student, and chairs the Kick-off, Green-light and Finalisation meetings. In that capacity, the delegate is there to protect the student against process errors and possibly unfair treatment by the

supervisors, while on the other hand ensuring that the student is assessed according to the appropriate criteria by the supervisors (Chapter 4).

3. Scientific approach

3.1. Research project

The graduation project is an individual research project that is conducted by the student with guidance and regular feedback from the supervisors.

The project starts with a preliminary phase where the student lays down the foundations for their research, which includes conducting a study of the most important relevant literature, refining their topic into a specific problem statement, establishing clear research questions or objectives, and selecting an appropriate methodology. This culminates in the creation of a graduation plan (see Appendix D), which is presented, discussed and assessed at the Kick-off meeting.

Upon approval of the graduation plan, the student conducts their research according to the methodology and time planning in it. The specific contents of this phase of the graduation will vary depending on the chosen methodology, but might include among others: an extended survey of scientific literature and other relevant sources, data collection and analysis, developing new methods, tools or datasets, and software development and testing.

The main deliverable of the graduation project is the written graduation report (see Appendix D), which documents the research that was conducted. It includes a reflection that is integrated into the report, mostly in the introduction and conclusions. The contents of the report will differ depending on the nature of the research, but they should be detailed enough to allow others to fully reproduce the work presented in it.

Similarly, in order to achieve this reproducibility, if the research involved the creation of source code or data, students are expected to publish these in a persistent public repository which is linked in the report. If the nature of the project does not allow for the full publication of the source code or data, the reasons for this should be documented in the report, and students are highly encouraged to publish code excerpts or synthetic data.

In addition to the graduation report, there are two other presentations as deliverables: (1) a technical presentation for the student's supervisors, which is given, discussed and assessed at the Green-light meeting, and (2) a technical presentation for an audience of fellow Geomatics students, which is given, discussed and assessed at the Finalisation meeting.

The graduation report and all presentations must be in English.

3.2. Use of AI and LLMs in the graduation project

The use of AI tools and large language models (LLMs) is generally allowed in the graduation project as a learning and research aid, for example for brainstorming, explaining concepts, checking grammar and style, reviewing the literature, or supporting data analysis and programming. However, AI should not replace learning or critical thinking: AI-generated output can be plausible-sounding but incorrect, biased or fabricated, so the student should always critically evaluate it. The student remains personally and fully responsible for all the work they submit, including anything produced with the help of AI, and must be able to explain,

defend and reproduce the substance of their work — and, for the thesis in particular, to independently defend all scientific content and contributions.

If AI has been used in any part of the project, the graduation plan and/or the graduation report must include an AI disclosure statement describing what tools were used, how they were used, and to what extent, along with a short reflection. The detailed requirements are set out in the Geomatics Policy for the use of AI/LLM, and the MSc Geomatics thesis template provides an appendix to help the student write the statement. Failure to disclose the use of AI is itself treated as a violation of the policy and may be regarded as fraud.

3.3. Graduation plan (Kick-off) structure

The graduation plan is a project proposal that must contain the following elements:

1. A **cover page** with
    1. title
    2. full name
    3. student number
    4. names of the responsible and second supervisors
2. An **introduction** in which the relevance of the project and its place in the context of geomatics is described, along with a clearly-defined problem statement;
3. A **related work** section in which the relevant literature is presented and linked to the project;
4. **Research questions** that are clearly defined, along with the scope (i.e. what the student will and will not be doing);
5. Overview of the proposed **methodology** to be used;
6. Overview of **preliminary results** (if applicable, to be discussed with the supervisors);
7. **Time planning** — a Gantt chart is generally preferable to a simple list;
8. Since specific **data and tools** have to be used, these should be presented concretely, so that the supervisors can see that the student has a grasp of all aspects of the project;
9. The **references**. Every reference must include a clickable DOI or URL, so that sources can be accessed directly from the document.

Supervisors expect a document of 10–15 pages.

A template is available in Appendix D.

If AI tools or large language models (LLMs) have been used for any part of the work behind the plan, the plan must include an AI disclosure statement, as described in Chapter 3.2.

3.4. Graduation report (Green-light and Finalisation) structure

The exact structure of the thesis, or graduation report, will depend on the type of work that is involved and should be discussed with the student's responsible supervisor.

We describe potential structures and provide templates at https://geomatics.bk.tudelft.nl/geo2022/templates/.

Notice that if AI tools or large language models (LLMs) have been used at any point in the project, the report must include a detailed AI disclosure statement, as described in Chapter 3.2.

4. Feedback and assessment

Throughout the graduation phase, the student should receive regular feedback from the supervisory team. More extensive feedback is provided during the Kick-off, Midterm and Green-light assessments.

The student submits their main deliverables — the graduation plan for the Kick-off and the MSc thesis report for the Green-light and Finalisation — one week before the corresponding assessment session, where they are then presented and discussed. The one-week deadline is the default: the responsible supervisor may agree with the student on a later submission deadline, as long as the supervisory team still has sufficient time to read the deliverables before the assessment. At the Kick-off, Green-light and Finalisation, the independent delegate chairs the session. Following the presentation, the student responds to the supervisory team's questions. Then, the supervisory team retreats for deliberation, after which the results are announced and feedback is provided.

During the Midterm assessment, the supervisors and student conduct a more structured meeting to discuss the progress and (updated) plans. The form of this session depends on the supervisors, but it might include a presentation, demo or extended discussion session.

At the Kick-off and Green-light, the result is expressed in a 'go' or a 'no-go'. In case of a 'no-go', a retake of the assessment can take place in the next quarter (see Chapter 1 and Appendix B).

4.1. Components of assessments

At the Kick-off, the graduation plan is assessed according to the Kick-off rubric (see Appendix E).

At the Midterm, the work is assessed based on whether it is quantitatively and qualitatively on schedule with respect to the graduation plan.

At the Green-light, the draft graduation report is assessed according to the Green-light and Finalisation rubric.

At the Finalisation, the final graduation report is assessed according to the Green-light and Finalisation rubric.

4.2. Deliberation and feedback

After the Kick-off, Green-light, and Finalisation sessions, the supervisory team will retreat for deliberation. At these sessions, the independent delegate chairs the deliberation (see Chapter 2 and Appendix B), without influencing the result.

During the deliberation, the supervisory team assesses the deliverables, including the presentation and the answers to the supervisory team's questions. At the Kick-off and Green-light, this assessment is formally expressed in a 'go' or a 'no-go', representing a passing or failing grade with respect to the corresponding rubric.

At the end of the deliberation for the Kick-off and Green-light, the responsible supervisor should communicate the results of the deliberation ('go' or 'no-go') and provide feedback to the student, including how to improve. At the Midterm, there might not be a deliberation,

but the supervisors should still provide feedback to the student. Afterwards, the responsible supervisor should register this result and feedback in MyCase.

The marks for the graduation project (GEO2022) are decided at the Finalisation milestone (see Chapter 1.6).

A. MyCase registration

As explained in Chapter 1, *MyCase* is used as the system to track the graduation progress of each student. All information about the graduation (general information and planning, stakeholders, documentation and assessment) is collected in this system.

The table below serves as a chronological overview of all tasks and responsibilities for students who start their thesis in Q2 (which is the standard/default case).

For example, if a student starts their thesis in Q3, adding +1 to the weeks mentioned in the table will produce the approximate timeline for the student. The submission of the topic should therefore be done before week 2.3 and the MyCase registration should be done weeks 2.8–3.2.

Phase	**When** (TU Delft week)	**What**	**Who**
Preparation	1.3	Submit thesis topic form in GEO2022 website	Student
	1.8–2.2	Register new case in MyCase, including the supervisors and the planning in weeks	Student
		Check entry requirements	Faculty administration
	2.5	Approve supervision team	Responsible supervisor
	2.6	Register delegate by BoE	MyCase faculty administration
	2.7	Schedule the Kick-off and register it in the GEO2022 website	Responsible supervisor
Kick-off	2.8–2.9	Upload graduation plan, human participation (see Appendix G), external party and confidentiality agreement, and submit ‘Ready for Kick-off’ task in MyCase	Student
	2.9–2.10	Kick-off assessment	Student, supervisors and delegate
		Submit result and feedback form in MyCase	Responsible supervisor
		Upload midterm materials, and submit ‘Ready for Midterm’ task in MyCase	Student
		Midterm assessment	Student and supervisors

Phase	**When** (TU Delft week)	**What**	**Who**
Midterm	3.8–3.9	Submit result and feedback in MyCase	Responsible supervisor
	4.2	Schedule the Green-light and register it in the GEO2022 website and register a date/ time slot for the Finalisation date in SuperSaaS*	Responsible supervisor
Green-light	4.3–4.4	Check study progress, upload draft graduation report, submit diploma application with final title and submit 'Ready for Green-light' task in MyCase, and upload draft graduation report in plagiarism scan in Brightspace	Student
		Check graduation requirements	Student programme administration
	4.4–4.5	Check for plagiarism	Responsible supervisor
		Green-light assessment	Student, supervisors and delegate
		Submit result and feedback in MyCase	Responsible supervisor
Finalisation	4.8–4.10	Upload finalisation presentation and submit 'Ready for finalisation' task in MyCase, upload graduation report and finalisation presentation in repository	Student
	4.9–5.1	Finalisation assessment	Student, supervisors, co-reader and delegate
		Upload final grade	Responsible supervisor

Information

*If a student does not follow the standard on-time path (Q2-Q4), the *SuperSaaS* tool might not be available. In such cases, the Finalisation should be planned by sending an email to graduation-bk@tudelft.nl.

B. Assessments

B.1. Preparation

Period	In fixed weeks, once per quarter
Actions student	• Find and agree on a topic with an MSc Geomatics staff member, who will act as supervisor • Register the student details, topic and supervisor in the GEO2022 website • Create case in MyCase

B.2. Kick-off

Admission requirements	The student must have obtained 45 ECTS from 9 out of the 10 mandatory courses, and have obtained 15 out of the 25 ECTS from the electives by the time of the Kick-off.
Period	In fixed weeks, once per quarter
Preparation	
Planning	Done by the responsible supervisor and registered in the GEO2022 website before the deadline in the Geomatics graduation calendar
Location	Room booked by the responsible supervisor
Deliverables	Graduation plan
Assessment method	Rubric
Actions student	• Upload graduation plan to MyCase (1 week before; unless the responsible supervisor agrees otherwise) • Prepare presentation
Actions responsible supervisor	• Book a room • Register planning in the GEO2022 website (date, time, room, student details, supervisors and delegate)
Actions supervisors	• Read the Graduation Plan before the start of the Kick-off
At the session	
Schedule	• 15 min: presentation by student, • 15 min: questions by second supervisor and responsible supervisor (in that order), • 15 min: deliberation and feedback
Assessors	Responsible and second supervisors
Quorum	Both supervisors and delegate
Actions responsible supervisor	Fill in the feedback form, upload it in MyCase and give feedback
Actions delegate	Chair the assessment and the deliberation

Upon completion	
Assessment result	• *Go*: the student proceeds towards the Midterm • *No-go*: the student agrees a new Kick-off date for the next quarter with the supervisors; registration is done by the responsible supervisor in the GEO2022 website • *Withdrawal*: same as for no-go.
Assessment registration	The feedback and assessment ('Go' or 'No-go') is registered in MyCase by the responsible supervisor

B.3. Midterm

Admission requirements	The student successfully passed the Kick-off milestone
Period	Anytime between Kick-off and Green-light, it is up to the student and the responsible supervisor to agree on a date
Preparation	
Planning	Done by responsible supervisor
Location	Room booked by the responsible supervisor
Deliverables	A document must be uploaded to MyCase. The form/ content of this document is decided by the supervisors.
Assessment method	(none)
Actions responsible supervisor	Book the room and invite the second supervisor
Actions student	Upload the required document to MyCase
At the session	
Schedule	Meeting has free form, it is decided by the supervisors
Assessors	(this is not a formal assessment)
Quorum	Both supervisors
Actions student	(none)
Actions supervisors	(none)
Upon completion	
Assessment result	(an advice is given to the student, based on their progress)
Assessment registration	The feedback/advice is registered in MyCase by the responsible supervisor

B.4. Green-light

Admission requirements	The student must have completed all Master's courses with the exception of GEO2022, and must have successfully completed the Midterm milestone.

Period	In fixed weeks, once per quarter.
Preparation	
Planning	Done by the responsible supervisor and registered in the GEO2022 website before the deadline in the Geomatics graduation calendar
Location	Room booked by the responsible supervisor
Deliverables	A draft thesis and a presentation
Assessment method	Rubric
Actions student	• Upload draft thesis to MyCase (1 week before; unless the responsible supervisor agrees otherwise), • Perform the plagiarism scan in Brightspace • Prepare presentation
Actions responsible supervisor	• Book a room • Register planning in the GEO2022 website (date, time, room, student details, supervisors and delegate) • Check results of plagiarism check
Actions supervisors	• Read the draft MSc thesis • Check if and how student implemented advice/feedback from the Midterm.
At the session	
Schedule	The times below are indicative; the assessors may divide the session differently, provided the assessment fits the booked time slot (see Appendix C). • 20 min: presentation by student • 20 min: questions by second supervisor and responsible supervisor (in that order) • 20 min: deliberation and feedback
Assessors	Responsible and second supervisors
Quorum	Both supervisors and delegate
Actions student	Give presentation and answer questions
Actions responsible supervisor	Fill in the feedback form, upload it in MyCase and give feedback
Actions delegate	Chair the assessment and the deliberation

Upon completion	
Assessment result	• *Go*: the student proceeds towards the Finalisation • *No-go*: the student agrees a new Green-light date for the next quarter with the supervisors; registration is done by the responsible supervisor in the GEO2022 website • *Withdrawal*: same as for no-go.
Assessment registration	The feedback and assessment ('go' or 'no-go') is registered in MyCase by the responsible supervisor

B.5. Finalisation (graduation)

Admission requirements	The student must have successfully completed the Green-light milestone.
Period	In fixed weeks, once per quarter. Must be done at least 4 weeks after the Green-light.
Preparation	
Planning	Day and time slot reservation (see Appendix C) made by responsible supervisor in SuperSaaS before the deadline in the Geomatics graduation calendar; if SuperSaaS is not available, by email to graduation-bk@tudelft.nl
Location	Room booked by the scheduling department
Deliverables	Final thesis and presentation
Assessment method	Rubric
Actions student	• Revise the thesis based on the feedback received at the Green-light • Upload final thesis to MyCase (1 week before the presentation; unless the responsible supervisor agrees otherwise) • Perform the plagiarism scan in Brightspace • Upload final thesis and finalisation presentation to the TU Delft repository (1 day before the presentation) • Prepare presentation
Actions responsible supervisor	• Book a date/time slot • Check results of plagiarism check

Actions supervisory team	• Read the final MSc thesis • Check if and how student implemented advice/feedback from the Green-light. • Send their tentative marks to the delegate before the session
At the session	
Schedule	• 20 min: presentation by student • 25 min: questions by co-reader, second supervisor, and responsible supervisor (in that order) • 10 min: deliberation • 10 min: private session with the student where the supervisory team explains the final grade • 10 min: laudation of the candidate and signing of the diploma
Assessors	Responsible supervisor, second supervisor, and co-reader
Quorum	Both supervisors, co-reader, and delegate
Actions student	• Give presentation and answer questions • Sign the diploma on both sides
Actions supervisory team	Fill in the final grade list
Actions supervisors	• Hand over the diploma and the grade list (the latter in a closed envelope) • Give the laudation (without announcing the final grade to the public)
Actions delegate	Chair the assessment and the deliberation
Upon completion	
Assessment registration	The final grade is registered in MyCase by the responsible supervisor
Actions student	• Unsubscribe as TU Delft student via Studielink enrolment, so your enrolment will be terminated as from the 1st of the next month (if you do not terminate your enrolment in time you are required to pay tuition fees for another month; retroactive termination of your enrolment is not possible; under certain circumstances the tuition fee can be partly refunded, see the website Termination of Enrolment) • Fill in exit evaluation (you will receive an email).

C. Time slots

The Kick-off, Green-light, and Finalisation assessments are scheduled in the standard time slots below, so that a room can be booked and the student, the supervisory team and the delegate can reserve the time in their agendas.

Kick-off	Green-light	Finalisation
08:45–09:30	08:45–09:45	08:45–10:00
09:45–10:30	10:00–11:00	10:15–11:30
10:45–11:30	11:15–12:15	11:45–13:00
11:45–12:30		
Break	*Break*	*Break*
13:45–14:30	13:45–14:45	14:45–16:00
14:45–15:30	15:00–16:00	16:15–17:30
15:45–16:30	16:15–17:15	
16:45–17:30		

The Midterm is not bound to these slots, as its date and form are agreed upon within the supervisory team (see Appendix B).

D. Deliverables

D.1. Kick-off template (graduation plan)

The graduation plan template (Kick-off) is available in two formats:

Typst https://typst.app/project/rxxc6SF4azqbGIOgdoRANf.
LaTeX https://gist.github.com/hugoledoux/d16d5a4d397858ac745e38f9e8561657.

Other formats such as Word or InDesign are also allowed.

D.2. Kick-off presentation

The Kick-off presentation is a 15-minute presentation of the graduation plan, giving a summary of the motivation, the problem, the research questions, the methodology and the time planning. The student can assume that the audience (the supervisory team) has read the graduation plan and is familiar with the problem and the research questions.

The goal of this presentation is to demonstrate that the plan is well-founded and feasible, so that the supervisory team can take the 'go' or 'no-go' decision on proceeding to the Midterm.

D.3. Final thesis template (Green-light + Finalisation)

There is no official template, and students are allowed to use the software and structure of their choice (after approval of their supervisor).

We recommend the following templates, but other formats such as Word or InDesign are also acceptable.

Typst https://github.com/tudelft3d/msc_geomatics_thesis_typst.
LaTeX https://github.com/tudelft3d/msc_geomatics_thesis_template.

D.4. Green-light presentation

The Green-light presentation is a technical 20-minute presentation of the main findings of the draft MSc thesis. It contains a technical summary of the entire graduation report, covering the research questions, methodology, results, and conclusions. The student can assume that the audience has read the draft MSc thesis and is familiar with the research questions and methodology.

The goal of this presentation is to demonstrate that the thesis work is complete and of sufficient quality to proceed to the final public defence (the Finalisation).

D.5. Finalisation presentation

In the Finalisation presentation, the student formally presents the complete findings of their MSc thesis to the supervisory team and fellow Geomatics students. The content is a technical summary of the entire graduation report, covering the research questions, methodology, results, and conclusions.

The presentation can be based on the Green-light presentation, but adapted to an audience of fellow Geomatics students who are technically proficient but have not read the thesis. The presentation lasts 20 minutes and is followed by a 25-minute question session.

The goal is to demonstrate the ability to communicate complex research effectively to a technical audience of fellow Geomatics students. After the Green-light, the student is expected to revise the thesis based on the feedback received and upload the final version to the repository before the Finalisation. The quality of this technical presentation contributes to the final grade, which is decided at the Finalisation.

E. Rubrics

E.1. Rubric for Kick-off

The graduation plan is assessed at the Kick-off with the rubric below. Each aspect is marked Go, Borderline or No-go; the pattern of marks is indicative, as the supervisors review where the aspects fall and make the final Go/No-go decision. A No-go on an essential aspect, or several Borderline/No-go marks, supports a No-go. The supervisors judge whether the student is on track to obtain a *Green-light* at the Green-light milestone.

Aspect	Go	Borderline	No-go
Motivation	Motivation is clearly stated and explicitly connected to the specific research problem and the context of geomatics.	Motivation can be broadly discerned, but is not clearly connected to the specific research problem.	No motivation is given, or the stated motivation does not match the proposed work.
Problem & research questions	The problem statement is clear with defined boundaries (scope) and is feasible; research questions are clearly defined with an explicit scope of what will and will not be done.	The problem or research questions are recognisable, but their scope or boundaries are not fully clear.	The problem cannot be explained, or no specific research questions or objectives are given.
Related work / literature	Relevant literature is presented and linked to the project, with adequate justification of the topic.	A sufficient introduction to the topic is given, but the literature review is limited or only weakly linked to the project.	The research is not placed in a wider context; literature is missing or superficial, and sources are accepted without reflection.
Methodology (methods / data)	The proposed methods and data are adequately justified and appropriate for the research questions.	Methods and data are partly justified, or their suitability for the research question is only weakly explained.	No adequate justification is given for the chosen methods and data, which are inappropriate for the research question.
Feasibility & time planning	A realistic time plan (e.g. Gantt chart) is given; the project is feasible within the graduation period and key risks are identified.	A time plan is given, but is optimistic or loosely linked to the work; risks are not clearly identified.	No real time planning is given, or the plan is not feasible within the graduation period.
Autonomy & use of supervision	Mostly autonomous and proactive; responds to feedback and contributes to meetings; implements suggested changes.	Sometimes autonomous, but generally needs steering; responds to feedback only minimally.	Not autonomous or proactive; constant steering required; does not respond to feedback or implement changes.
Structure & writing	Plan follows a clear structure with the required elements (introduction, related work, research questions, methodology, time planning, data & tools, references) and is generally well written with few errors.	Plan follows a structure but with some issues in clarity/organisation; writing has a number of errors needing correction.	Plan has no clear structure or logical flow; writing is disorganised with pervasive errors that obscure meaning.
References & AI disclosure	Other work is acknowledged properly with a complete, consistent reference list; if AI/LLMs were used, a disclosure statement describes tools, use and extent.	References are present but incomplete or inconsistent; or AI use is disclosed only partially.	Sources are not acknowledged or references missing/unreliable; or AI/LLMs were used without the required disclosure statement.
Content & structure	Presentation follows a clear structure and gives a good summary of motivation, problem, research questions, methodology and planning.	Presentation follows a structure but with minor issues; summary covers most key elements.	Presentation is chaotic or does not convey the motivation, problem or main elements of the plan.
Delivery & visual material	Adequate presentation material supports the talk; interaction with the audience is appropriate and maintains attention.	Basic but functional material; interaction with the audience is sufficient though not always maintained.	Visual material is missing or of poor quality; loses the audience rapidly.
Understanding & Q&A	Candidate answers most questions correctly and is confident with the content for its application; understands the plan and its implications.	Candidate answers some questions but with gaps or superficial reasoning when probed.	Candidate cannot address the questions posed or demonstrates a clear lack of understanding of the problem.

E.2. Rubric for Green-light and Finalisation

The rubric below lists the aspects against which the draft report and its presentation are assessed at the Green-light and the Finalisation. The categories are weighted: Research (50%), Process (20%), Communication (30%) of which Report (18%) and Presentation (12%). The supervisors assess all categories, while the co-reader assesses the Research and Communication categories only (see Chapter 2). One grade is given per category; the aspects serve as a checklist to support the judgement. Half-point grades may be given; the descriptors are formulated at whole grades, and a half-point grade is judged between the two neighbouring descriptors. Grades of 5.75 and higher round up to 6 and count as a pass. *(Digital version in PDF and XLSX is available at https://geomatics.bk.tudelft.nl/geo2022/rubric/;)*

Aspect	< 5.75	6	7	8	9	10
Research (50%)						
Motivation	No motivation is given, or the stated motivation does not match the work.	Motivation can be broadly discerned, but is not well understood.	Motivation can be understood and related to the problem.	Motivation is clearly stated and explicitly connected to the specific research problem.	Motivation is clearly argued and connected to the wider need for solutions of the problem.	Motivation is argued convincingly, grounding the research in its wider scientific and practical context and making clear why the problem matters.
General problem	The general problem cannot be explained; no specific research questions or objectives are given.	The general problem is recognisable, but its scope and boundaries are not fully clear.	The general problem is clear with defined boundaries (scope).	The general problem is clear and has defined limitations.	The general problem is clear, has boundaries or limitations and is feasible.	The general problem is clear, has boundaries or limitations and is feasible with the approach proposed.
Theoretical framework	No relevant theory is reproduced or applied to the research.	Understands directly relevant theory at MSc level, but has difficulties applying it to the research.	Understands directly relevant theory and applies it to the research after being shown how.	Understands and can reproduce directly relevant theory at the level of MSc textbooks and scientific literature, and applies it to the research.	Independently collects, processes and integrates theory from different fields or sources, and applies it to the research.	Independently integrates and extends theory from different fields or sources, making a clear conceptual contribution to the research.
Literature / related work	Unable to place the research in a wider context; no clear literature research; sources are accepted without reflection.	Sufficient introduction and justification of the topic, but superficial (limited literature review).	Sufficient introduction and justification of the topic, with fair literature support (decent literature review).	Good introduction and justification of the topic with supporting literature (but not all included).	Good introduction and justification of the topic, with vast literature support and critical evaluation of sources.	Excellent introduction and justification of the topic, with all literature support, including recent related work by other researchers.
Choices of methods / data	No adequate justification is given for the chosen methods and data, which are inappropriate for the research question.	Choices of methods and data are partly justified, or their suitability for the research question is only weakly explained.	Choices of methods and data are adequately justified and appropriate for the research question at hand.	Choices of methods and data are justified, appropriate and logical, with only minor gaps in reasoning.	Choices of methods and data are justified, logical and well-matched to the research question, with alternative approaches considered.	The choices of methods and data are justified, logical and the most efficient at the moment.
Results / conclusions	No substantial conclusions; results are left uninterpreted.	Results are interpreted to a limited extent.	Results are interpreted independently with a critical attitude.	Results are interpreted critically and reflected upon within the broader scope of the discipline and its application.	Beyond a critical, discipline-wide interpretation, the work proposes solutions or alternative approaches where the evidence is weak, showing how the results inform practice.	The work offers a critical, discipline-wide reflection on the results and puts forward well-argued solutions or alternatives with clearly stated implications for future research and application.
Answers to research questions	The results do not answer the research questions.	The answers to the research questions are only partial or indirect; one or more questions are left unaddressed or answered tentatively.	Each research question is addressed at a basic level, with limited depth or supporting evidence.	The research questions are answered clearly and with supporting evidence, though minor gaps remain for one or two questions.	The research questions are answered thoroughly and with strong evidence, with claims well supported across all questions.	The research questions are answered comprehensively and convincingly, leaving no significant gaps and linking the answers back to the stated motivation.
Depth & ambition of the investigation	The topic avoids meaningful challenge and the treatment is superficial; little genuine investigation is attempted.	The topic is modest in scope and the treatment is largely surface-level, with limited independent investigation.	The topic is appropriate for an MSc and is investigated to a satisfactory depth; the student engages the core problem without much extension.	A moderately challenging topic taken to good depth, or a simpler topic explored thoroughly; the investigation goes beyond the obvious.	A challenging topic investigated deeply, or a simpler topic taken to notable depth with rigorous analysis; clearly beyond minimum requirements.	A demanding topic pursued with exceptional depth, or an apparently simple topic transformed through rigorous, thorough investigation; the level of inquiry is exemplary.
Process (20%)						
Autonomy / proactiveness	Not autonomous or proactive at all; constant steering by supervisors is required.	Sometimes autonomous and proactive, but generally needs steering by supervisors.	Mostly autonomous; generally tries approaches before asking for help.	Mostly autonomous and proactive, taking control of the project and steering it to completion with some hiccups.	Autonomous and proactive, taking control of the project and steering it.	Highly autonomous and proactive throughout, taking full control of the project and steering it efficiently.
Response to feedback / meetings with supervisors	Does not respond to feedback or suggested alternatives; required changes are not implemented.	Responds to feedback only minimally; implements few changes and shows little improvement.	Contributes to discussions during meetings; critical attitude, but most key issues had to be pointed out by supervisors; uses feedback.	Contributes to lively discussions; critical attitude, but key issues had to be pointed out; uses feedback for self-improvement.	Leads lively discussions; critical attitude, pointing out the issues themselves; uses feedback for self-improvement.	Leads lively discussions; critical own attitude; actively uses both own discoveries and feedback for self-improvement.
Use of resources	Misuse of resources (data, computational time, people time).	Makes inefficient but passable use of resources (e.g. tools, data, own/supervisor's time).	Use of resources is appropriate (e.g. tools, data, own/supervisor's time).	Makes good use of resources (e.g. tools, data, own/supervisor's time).	Makes very good use of resources (e.g. tools, data, own/supervisor's time).	Makes highly efficient use of resources (e.g. tools, data, own/supervisor's time).
Originality / creativity	No original ideas within the project; most of the work is copied or already developed.	Contribution to the project is somewhat original; limited initiative and suggestions within the project.	Contribution to the project is partly original; some initiative and suggestions by the student.	Contribution to the project is original, with suggestions by supervisors; several initiatives within the project.	Contribution to the project is original, with almost no intervention by supervisors; many initiatives within the project.	Contribution to the project is original; always takes initiative and makes suggestions within the project.
Planning	No real planning; missed most of the deadlines.	Basic timeline and plan prepared, but little followed or updated.	Good timeline and plan prepared, often followed or updated.	Prepared a good and feasible plan, mostly followed or adjusted when needed (e.g. according to progress and new findings).	Prepared a clear and feasible plan, consistently followed and actively improved in response to progress and new findings.	Prepared an efficient, clear and feasible plan, consistently followed and improved, managing changes smoothly and keeping the project on schedule.
Comm - Report (18%)						
Structure	Report has no clear structure or logical flow.	Report follows a structure, but with significant issues in clarity and organisation.	Report follows a structure, with some issues in clarity or organisation that do not seriously impede understanding.	Report follows a clear structure, with only minor issues in clarity.	Report follows a clear and logical structure throughout.	Report follows a clear and logical structure in which every section serves the argument and guides the reader.
Documentation of work done	The report omits key parts of the research, so the process and results cannot be understood.	The report documents the research only partially; important steps are missing or unclear.	The report documents all main parts of the research, so the process can be followed.	The report documents all parts of the research clearly, including data handling and key decisions.	The report documents all parts of the research in detail, including data handling, decisions and assumptions.	The report documents every part of the research in detail, giving a complete and faithful account of the process.
Writing	Writing is disorganised, with pervasive errors that obscure meaning.	Report is written with limited clarity and contains significant errors that need correction.	Report is generally well written, but contains a number of errors and needs improvements.	Report is generally well written, but contains a few errors and needs improvements.	Report is well written, with very few writing errors.	Report is well written using clear scientific language, with few errors.
Abstract	Abstract is missing or fails to summarise the work.	Abstract captures little of the work, or is unbalanced in what it covers.	Abstract captures the main elements of the work, with some gaps in coverage.	Abstract captures most of the work, including the main methods and results.	Abstract captures the essence of the work, covering aim, method and main results.	Abstract captures the essence of the work in a concise and complete summary.
Use of references	Sources are not acknowledged; references are missing or unreliable.	Other work is acknowledged to a limited extent; the reference list is incomplete or inconsistent.	Report properly acknowledges other work broadly and contains a fair list of references.	Report properly acknowledges other work most of the time, with a mostly complete reference list.	Report properly acknowledges other work consistently, with a complete reference list and only minor formatting issues.	Report properly acknowledges other work everywhere and contains a complete and well-formatted reference list.
Supplementary output / data	No supplementary output or data is provided.	Supplementary output or data is mentioned, but is broken, incomplete or unusable.	Work yields limited but usable supplementary output (e.g. software, data).	Work yields useful supplementary output (e.g. software, data), which is added to the report.	Work yields substantial supplementary output, which is added to the report and made available to the reader.	Work yields substantial, well-documented supplementary output (e.g. software, data), which is made publicly available alongside the report.
Reproducibility / open science	The research process cannot be reproduced or verified.	Reproducibility is poor; code, data or parameters are missing or undocumented.	The research is mostly reproducible; code and data are documented at a basic level.	The research is reproducible; code and data are organised and documented.	The research is reproducible following good open science practices (e.g. version control, documentation).	The research is fully reproducible by others: code, data and parameters are complete, documented and publicly available following open science best practices.
Comm - Presentation (12%)						
Structure	Presentation is chaotic; structure not clear.	Presentation follows a structure, but with significant issues in clarity and organisation.	Presentation follows a structure, with some issues in clarity or organisation.	Presentation follows a clear structure, with only minor issues.	Presentation follows a clear structure with good pacing and transitions.	Presentation follows a clear, logical and well-paced structure that leads the audience through the argument.
Content	Presentation does not convey the motivation, problem or main results of the work.	Presentation gives a partial summary of the work; some key points are missing or unclear.	Presentation gives a decent summary of motivation, problem, work done, results and conclusions.	Presentation gives a good summary of motivation, problem, work done, results and conclusions.	Presentation gives a very good summary of motivation, problem, work done, results and conclusions.	Presentation gives a concise, complete and easy-to-follow summary of motivation, problem, work done, results and conclusions.
Visual material	Visual material is missing or of poor quality.	Basic presentation material (e.g. slides, videos, demos), functional but plain.	Adequate presentation material that supports the talk.	Good presentation material, with clean visuals that support the talk.	Very good presentation material, with well-designed visuals.	Excellent presentation material, with clear, well-designed visuals that strengthen the talk.
Audience / attention	Loses the audience rapidly.	Interaction with the audience is sufficient (eye contact, body language, tone of voice, pace of speaking); gets the attention of the audience.	Interaction with the audience is appropriate; gets the attention of the audience and maintains it to some extent.	Interaction with the audience is good; maintains the attention of the audience for most of the presentation.	Interaction with the audience is very good; maintains constant attention of the audience.	Interaction with the audience is excellent; maintains constant attention of the audience throughout.
Responses to questions	Candidate cannot address the questions posed.	Candidate answers some questions correctly, but answers are superficial or incomplete.	Candidate answers most questions correctly, with a few gaps.	Candidate answers questions correctly, with a clear explanation of the reasoning.	Candidate answers questions correctly and responds well to follow-up questions.	Questions are answered succinctly and with full awareness of the strengths and weaknesses of the research.
Understanding	Demonstrates a clear lack of understanding of the scientific problem and cannot explain the work.	Shows superficial knowledge of the topic; struggles when probed beyond the slides.	Explains the work competently for its intended application; handles the main questions but shows occasional gaps when probed on details.	Explains the work confidently at a research-and-development level; handles most probing questions without hesitation.	Masters the content of the research topic; answers probing questions fluently and can articulate the work's limitations.	Masters the content well beyond the immediate research topic; situates the work in the wider field and discusses limitations and implications unprompted.

F. Particular circumstances

F.1. Quorum at formal assessments

A quorum is required for each assessment to be valid; it is as follows:

	Responsible supervisor	Second supervisor	Co-reader	Delegate BoE
Kick-off	✅	✅		✅
Midterm	✅	✅		
Green-light	✅	✅		✅
Finalisation	✅	✅	✅	✅

In case of absence, action will be taken as follows:

Absence of the delegate of the Board of Examiners The Board of Examiners appoints the delegate of the Board of Examiners and the substitute delegate of the Board of Examiners for all students. If the delegate of the Board of Examiners cannot attend an assessment, the substitute will be asked and the Secretary of the Board of Examiners will be informed on this replacement. The substitute delegate of the Board of Examiners is registered in MyCase.

Absence of one of the supervisors and/or the co-reader If it is known in advance that a supervisor or the co-reader will be unable to attend, a presentation must be held for that assessor prior to the evaluation. The assessment of that assessor must be written down with comments and feedback. This assessment must be given to the delegate of the Board of Examiners in a signed letter within a closed envelope or sent by e-mail. At the deliberation, this assessment will be taken into account by the other relevant members of the supervisory team for determining the final assessment.

In case of an unexpected absence, the other supervisory team members will search for a replacement. The Secretary of the Board of Examiners is also informed by the responsible supervisor or the delegate of the Board of Examiners about this absence. The assessment should preferably be continued, but the final assessment should be determined after hearing the absent supervisor.

The determination for a ‘go’ or ‘no-go’ at the Kick-off and Green-light can only take place after consulting the absent assessor. If this is not possible, the final assessment is postponed. In both cases, a meeting with the absent supervisor takes place on the shortest possible term, to determine a final conclusion. In case of doubt or on request of the student, it may be decided that an extra assessment must be held.

F.2. Difficulties at the deliberation

It may occur that the deliberation does not lead to an unambiguous assessment. The delegate of the Board of Examiners informs the student on this situation and explains the applied procedure and the corresponding terms. Subsequently, the presented products will be handed over to the Secretary of the Board of Examiners. The Secretary of the Board of

Examiners decides which member of the committee will look after this matter. This member of the Board of Examiners will reconvene the supervisory team and the delegate of the Board of Examiners for a redeliberation, chaired by the member of the Board of Examiners, in which an attempt is made to reach consensus. If unexpectedly this does not succeed, the member of the Board of Examiners will make a final decision.

F.3. Plagiarism

The plagiarism scan on graduation documents is implemented for all graduating students, to guarantee the authenticity of the graduation work. For this, the **Turnitin** tool in Brightspace is used. The student can upload provisional versions of their document as often as desired to check them for originality. One week before the Green-light and Finalisation presentations, the draft and final versions of the MSc thesis must be uploaded by the student in order for them to be scanned. It is the responsibility of the responsible supervisor to determine whether the results of the plagiarism scan are an indication of actual plagiarism. If there is a suspicion of plagiarism, the responsible supervisor should share the findings with the student, the second supervisor and co-reader at the Green-light and Finalisation. If there is a suspicion of intentional plagiarism, the responsible supervisor should discuss this with the student and notify the Board of Examiners afterwards.

In addition to the plagiarism scan, the coordinator of the Geomatics Graduation Studio performs a check for fake/hallucinated references on all graduation reports. The student does not need to take any extra action: the check runs on the deliverables that are already submitted to MyCase. It is then the responsibility of the responsible supervisor to review the findings and discuss them with the student, the second supervisor and co-reader if needed, following the same procedure as for the plagiarism scan.

Remember that all use of LLMs/AI tools must be clearly documented in the student's thesis, see Chapter 3.2.

G. Data management checklist

This checklist is relevant for all graduation projects at the Faculty of Architecture and the Built Environment. The form is intended to highlight common aspects of graduation projects that require particular attention with regard to planning the research and data management. Relevant information and supplementary sources regarding each question are provided below each question.

With this checklist, the faculty wants to prevent students from unexpectedly finding themselves in complex and stressful situations, in which ethical or privacy matters and/or other laws and regulations become an issue. In projects involving humans, certain types of data processing increase the risks to the human participants. Planning such projects requires additional evaluations and advice from university staff before ethical approval can be received and the project can begin. In the case of a graduation project, obtaining additional advice or permits may delay the project with an extra education period or semester. To avoid this, it is recommended that students set up a graduation project with a low level of risk. Therefore, all students have to check their risk, by completing this checklist before their Kick-off.

The first section of the checklist (A) should be completed by all students, together with their supervisor, during the planning of the graduation project, before the Kick-off. It does not need to be submitted to anyone for review or approval. Please consider questions 1 to 3 carefully in relation to the intended graduation project, and answer with 'yes' or 'no'.

The second section of the checklist (B) should only be completed if the graduation project involves working with data from human participants. In that case, the student and their supervisor must apply for and receive ethical approval from the Human Research Ethics Committee (HREC) before the project can begin (see the paragraph 'Explanation and follow-up' after the questions). The student can submit the application to the HREC, but the supervisor is responsible for making sure that the project is compliant with relevant privacy regulations and ethical policies.

	Section A. General considerations	**yes**	**no**
1	Is the graduation project conducted as part of an internship (at a company), or as part of a research project at TU Delft? If a student's graduation project is conducted at a company or as part of a research project at the university, questions of data ownership and intellectual property rights need to be addressed in a written graduation or internship agreement before the project begins. Students and their supervisor should consult the Intellectual Property Rights of Students webpage. Additional information can also be found in the Extended Personal Research Data Workflow. If applicable, complete the Confidentiality Agreement (this must be discussed with both supervisors first).		
2	Does the project involve conducting (part of) the research outside the Netherlands?		

	Section A. General considerations	**yes**	**no**
	Students who intend to travel abroad (even to other EU countries) for study, exchange, research, internship, or graduation project purposes need to follow the Travel Safety Protocol. This includes attending a mandatory Travel Safety Training Session: see the Disclaimer		
3	Will the research involve processing data from humans, such as running a survey, conducting interviews or workshops, collecting data through social media or internet forums, or re-using existing datasets about humans provided by a third party? (If 'yes', see follow-up questions 4 to 13 in Checklist B.) Students who work with data from human participants must complete the next section and apply for and receive ethical approval from the Human Research Ethics Committee (HREC) before conducting the research.		

	Section B. Extended risk factors (only if question 3 has been answered with 'yes'.)	**yes**	**no**
4	Will the project involve participants who may be considered vulnerable, such as the elderly, refugees or asylum seekers, ethnic minorities, patients, or people with disabilities? Participants who may suffer very adverse consequences (for instance, due to discrimination) if their personal data became publicly available can be considered vulnerable.		
5	Will the project involve participants who cannot themselves give informed consent for taking part in the project, but for whom consent must be obtained from a legal guardian? Participants who cannot give informed consent can include, for instance, children or participants with intellectual disabilities, mental disorders, or dementia. Such participants are also considered vulnerable in the context of the General Data Protection Regulation (GDPR).		
6	Will the project involve processing any of the special categories of personal data below? • Race • Ethnicity • Criminal offence data • Political opinion • Union membership • Religious or philosophical beliefs • Sex life and/or sexual orientation • Health data (including measurements such as heart rate)		

	Section B. Extended risk factors (only if question 3 has been answered with 'yes'.)	**yes**	**no**
	• Biometric or genetic data (including fingerprints, iris scanning, facial recognition) The General Data Protection Regulation (GDPR) defines stricter rules for processing special categories of personal data. If it is necessary to process these data in a project, it is important to provide additional safeguards.		
7	Will the project involve processing personal data that could be considered sensitive, such as the ones listed below? • Information about a person's income, debts, or other payments • Information about a person's (un-)employment status • Information about a person's performance at school or work • Information about relationship problems or (gambling) addiction • Information about poverty, domestic violence, or youth welfare/ social work involvement Some types of personal data are considered sensitive, because they can have a high impact on the privacy of the data subject if other persons gain access to these data. Sensitive personal data should only be processed if necessary: in such cases, additional safeguards need to be put in place.		
8	Will the project involve processing video-recordings, or photographs of participants? TU Delft considers photographic and video-materials of research participants to be sensitive personal data. If such data need to be processed, additional safeguards must be put in place.		
9	Will the project involve sharing or transferring personal data between multiple partners or collaborating organisations involved, such as between TU Delft and an internship company? According to privacy law, sharing personal data between organisations requires a privacy agreement to be in place: setting this up takes time, and requires support from additional university staff. Furthermore, personal data sharing can potentially expose research participants to different types of risks: these risks must be considered in the ethical application.		
10	Will the project involve deception, or covert observation of participants?		

	Section B. Extended risk factors (only if question 3 has been answered with 'yes'.)	**yes**	**no**
	In some types of research, obtaining informed consent for processing participants' personal data is not an option: for instance, if the research involves deception, or the research is covert (conducted without participants knowing about it). In such situations, the steps to mitigate risks to participants are important, and an alternative legal basis for processing the participant's data needs to be established with the help of additional support staff.		
11	Will the project involve working with social media data? Social media data are personal data, but since it is usually not possible to ask for informed consent for processing social media data, another legal basis for processing the participant's data needs to be established. Processing of social media data also involves legal considerations related to terms of use of data from third-party platforms: therefore, research with social media data requires expert support on privacy, ethics, and legal matters.		
12	Will the project involve using learning algorithms or other AI to analyse, combine, or otherwise process data from participants? The use of AI in research involves many considerations in terms of data protection, ethics, security, and intellectual property: for more information, see TU Delft's Instructions for use of Generative AI.		
13	Will the project involve participants who are based in a country or countries outside of the EU? Students affiliated with TU Delft must comply with Dutch and EU regulations of personal data processing (GDPR). Furthermore, the student and their supervisor must make sure that the research complies with local (privacy) legislations of any foreign destinations. Additional support from an external (local) expert may be required.		

Explanation and follow-up

If you have answered 'no' to all questions 4 to 13, your project is likely to be considered low or minimal-risk: see the paragraph 'Projects with minimal or low-risk' on the next page.

If you have answered 'yes' to one or more of the questions 4 to 13, your research likely involves extended or high risks to participants, according to the General Data Protection Regulation (GDPR) and TU Delft's privacy and ethical policies: for information regarding such projects, see the paragraph 'Projects with extended or high-risk' on the next pages.

Projects with minimal or low-risk

If you have answered 'no' to questions 4 to 13, your project is likely to be considered low-risk. This does not mean that the project involves no risks at all, but suggests that these risks can likely be addressed by the student and supervisor in the application to the Human Research Ethics Committee (HREC) within the timeline for a graduation project and without need for additional support.

Compiling the HREC application:
An application to the HREC generally involves a Data Management Plan (DMP), a risk-identification and mitigation checklist, and informed consent materials. Master's students at ABE who intend to compile a HREC application are advised to make use of the following support documents:

- the student guide
- the Example Data Management Plan for MSc projects

The graduation supervisor is responsible for the student's project and ethical application, and must provide support for compiling the HREC application documents.

Additional support
For low-risk student graduation projects, compiling of the HREC application documents should be done by the student in consultation with the supervisor. The Faculty Data Steward can be contacted for individual questions at datasteward-BK@tudelft.nl; however, the Data Steward does not provide detailed feedback on student DMPs for low-risk HREC applications.

Additional resources

The HREC has guides available for completing the checklist and for compiling informed consent materials.

Additionally, the Guide to the Extended Personal Research Data Workflow has been created to help researchers and students who work with human participants comply with both GDPR principles and TU Delft's policies on Data Management and Human Research Ethics.

Timeline

Minimal or low-risk HREC applications are generally processed faster than extended or high-risk applications (see the paragraph below). Nevertheless, the initial evaluation by the HREC usually takes approximately 2 weeks, and may take longer during busy periods or holiday: see the HREC website for up-to-date information. Additionally, the application may require revisions before final approval is granted. If you do not receive an initial response about your ethical application after 4 weeks from the time of submission, you may follow up with the HREC to enquire about an update.

Projects with extended or high-risk

If you have answered 'yes' to one or more of questions 4 to 13, there are potential increased risks related to how data from human participants will be processed in your project. These risks will need to be addressed in consultation with the Data Steward and other relevant

support staff before submitting the ethical application to the Human Research Ethics Committee (HREC)

Compiling the HREC application

An application to the HREC generally involves a Data Management Plan (DMP), a risk-identification and mitigation checklist, and informed consent materials. Master's students at ABE who intend to compile a HREC application are advised to make use of the following support documents:

- the student guide
- the Example Data Management Plan for MSc projects

The graduation supervisor is responsible for the student's project and ethical application, and must provide support for compiling the HREC application documents.

Additional support

Once the DMP has been compiled and reviewed by the supervisor, feedback should be requested from the Data Steward via DMPonline. After this, any other necessary support staff will need to be contacted. Crucially, if the project involves one or multiple ways of personal data processing that could result in high-risk to the participants according to the GDPR, the TU Delft Privacy Team must be consulted to establish whether or not a Data Protection Impact Assessment (DPIA) is required.

Additional resources

The HREC has guides available for completing the checklist and for compiling informed consent materials. Additionally, the Guide to the Extended Personal Research Data Workflow has been created to help researchers and students who work with human participants comply with both GDPR principles and TU Delft's policies on Data Management and Human Research Ethics.

Timeline

It can take a long time to compile a complete research plan and HREC application for projects involving extended risks. DMP feedback from the Data Steward usually takes around 2 weeks, but can take longer during busy periods or holidays. Receiving additional support from other staff, such as the Privacy Team, can take anywhere from a few days to multiple weeks, depending on the project and capacity of university staff. If a DPIA is deemed necessary, it can take anywhere from 4 weeks to several months.

It is important to note that advice from the Privacy Team or other support staff, as well as any additional documents (such as necessary contracts, or a DPIA, if needed) must be in place before the application is submitted to the HREC. The initial evaluation by the HREC can be processed in 2 weeks, but may take longer during busy periods or holidays: see the HREC website for up-to-date information. Additionally, the application may require revisions before final approval is granted. If you do not receive an initial response about your ethical application after 4 weeks from the time of submission, you may follow up with the HREC to enquire about an update.

Considering the limited time available for students conducting their graduation projects, students working with data from human participants are strongly advised to prioritise low-risk research projects. If a student project necessitates processing data in ways that are considered extended or high-risk, both student and supervisor need to be aware of the extended processing times involved in obtaining ethical approval and beginning the graduation project.

H. Final attainment levels

H.1. Generic Final Attainment Levels (from TU Delft)

A student graduating with a Master's in Geomatics:

G1	**Is skilled in one or more academic disciplines.** A university graduate is familiar with existing academic knowledge and is capable of expanding this through study.
G2	**Is able to perform research.** A university graduate is capable of acquiring new academic knowledge through research. In this connection, 'research' means developing and discovering new knowledge and insights in a purposeful and methodical manner.
G3	**Is able to design.** Many university graduates will formulate designs in addition to performing research. Designing is a synthetic activity geared towards bringing about new or modified artefacts or systems, for the purpose of creating values in accordance with the requirements and wishes stated beforehand (for example, mobility or health).
G4	**Has an academic approach.** A university graduate follows a systematic approach, characterised by developing and using theories, models and coherent interpretations, has a critical attitude and has an understanding of the unique nature of science and technology.
G5	**Possesses basic intellectual skills.** A university graduate is able to reason, reflect and form judgements. These are skills which are learned or sharpened in the courses of a discipline, and are generically applicable afterwards.
G6	**Is able to cooperate and communicate.** A university graduate has the capacity to work with and for others. This requires not only sufficient interaction, a sense of responsibility, and leadership, but also good communication with individuals both in and outside the profession. In addition, the university graduate is able to participate in academic or public debates.
G7	**Takes into account the temporal and social context.** Science and technology are not isolated, but rather, always have a temporal and social context. Opinions and methods have a background; decisions have social consequences over time. A university graduate realises this and is able to integrate these insights into their academic work.

H.2. Specific final attainment levels for the Master's Geomatics

A graduate of the Master's programme in Geomatics:

GM1	Is familiar with existing scientific knowledge, and has the competence to increase and develop this through study, and is competent in one or more scientific disciplines covering the complete geo-information chain, by:

	a) having appropriate knowledge of the role of geo-information in decision making applied to the built environment b) applying a formal, theoretical and methodologically sound approach to store, process, manage, disseminate and visualise geo-data and geo-information c) acting at the forefront of data analysis methodology and spatio-temporal database management systems technology d) having appropriate knowledge of legal and organisational aspects of geo-information; and e) utilising a technology and application driven approach to geo-data acquisition and geo-information extraction.
GM2	Has the research competences to: a) investigate and analyse spatio-temporal models and systems of complex real-world situations b) formulate hypotheses or research questions and develop new concepts c) validate developed concepts by appropriate prototypes, and d) analyse and evaluate thoroughly obtained results.
GM3	Has the competences to design, implement and use: a) models and systems for management, storage, processing, integration, exchange and dissemination of geo-information b) systems and methodologies to analyse measurements and other evidence material for the purpose of geo-information extraction; and c) select, combine and extend above-mentioned systems and methodologies to meet geo information requirements, notably in the built environment, in the present and in the future.